LAB 3

OSPF - Open Shortest Path First & NAT – Network Address Translation

Objective

In this lab you will learn how to implement and configure OSPF routing processes. You will also learn how to configure NAT. The lab has been divided into six sections. Each section will emphasize on some fundamental concepts of OSPF, starting from basic to the more complicated. The last section will look at implementing NAT overloading, also known as Port Address Translation (PAT) in the OSPF enabled network.

After completing this lab you will know how to:

- Enable OSPF on routers and specify the range of the associated IP addresses
- Configure OSPF Area Parameters over Serial Interfaces
- OSPF point-to-point network configuration
- Create a virtual link to the backbone area
- Configure NAT overloading

Background

OSPF (Open-Shortest Path First) is a link-state protocol where each router advertises the states of its neighboring routers to all other routers in its area. OSPF router uses LSA (link-state advertisements) to disperse routing information. As OSPF routes accumulate link-state information, it uses the Dijkstra algorithm to calculate the shortest path with the smallest cost to each node. All routers will exchange LSAs by means of flooding. LSA updates are sent only if a change occurs, such as router or link goes down, or a network being added.

OSPF routing process divides the network into autonomous systems (AS), which are a collection of networks under a common administration and routing strategy. In turn, AS is divided into many areas. Each AS has a backbone area, or area 0, to which all other area are attached (see figure 1 below). The OSPF backbone is responsible for distributing routing information between areas. OSPF is interior which means it exchanges LSA updates between routes within the same area.

This network hierarchy results in two different types of OSPF routers and link-state packets. A router that has all its interfaces within the same area is called internal router (IR). Routers within the same area have the same exact link-state database. Routers with interfaces in multiple areas are called area border router (ABR) and are required to transmit routing information and changes between areas (example router R10, and R15 in the figure.) Finally, a router that can provide routing information external to the AS is called autonomous system border router (ASBR). ASBR can perform redistribution between OSPF and other routing protocols such as RIP.

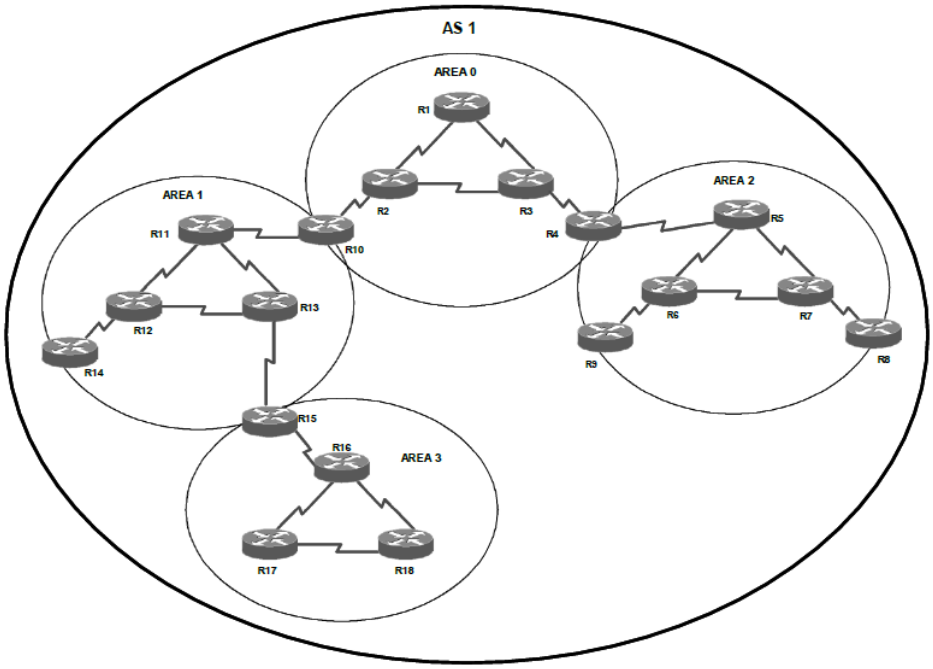


Figure 1. OSPF, Areas of Autonomous System

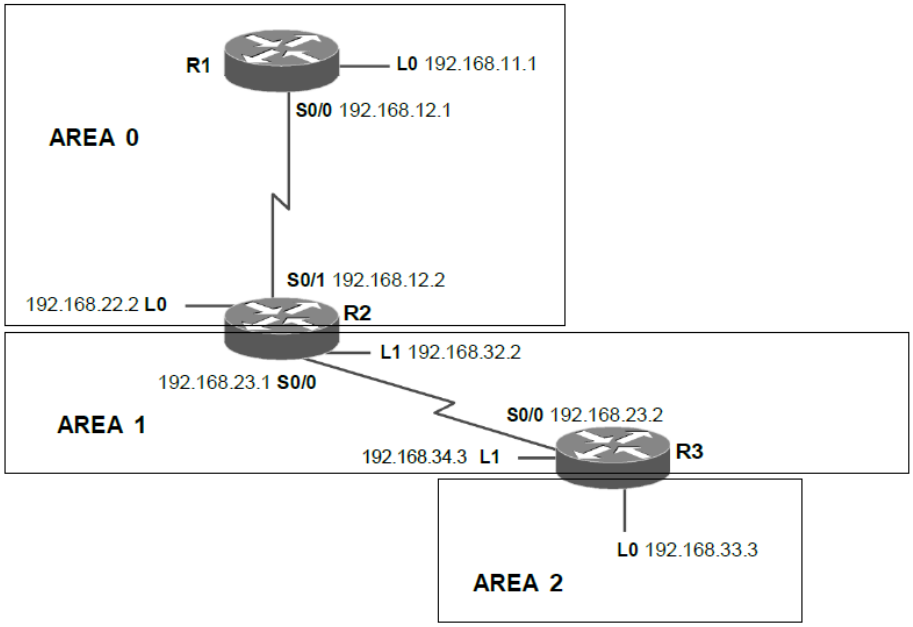


Figure 2 - OSPF Topology

Network Topology

To perform the different exercises in this lab you will be using the network topology depicted in Figure 2.

Experiments

Configuring OSPF
To configure the router for an OSPF routing process you need be at the global configuration mode, then use the commands:
router ospf <*process id*> *enable OSPF routing,*
Placing you in router configuration mode

network <*ip address*> <*wildcard mask*> **area** <*area id*>
Defines an interface on which OSPF runs and defines the area ID for that interface

The process id is used internally by the router which can be any integer. Multiple OSPF routing processes using the same process id # can be configured, but it's not recommended. The "network ip address" is the address of directly connected networks or subnets. The wildcard mask is an inverse mask that compares a given address with interface addressing to determine whether OSPF uses this interface. For example, 0.0.255.255 indicates a match in the first 2 bytes of the network number, 0.0.0.0 indicates a match in all 4 bytes of the network number. The area parameter identifies the interface as belonging to an area. The area id specifies the area associated with the network address.

EXERCISE 1: Configuring Routers with OSPF

STEP 1: Configure IP addresses for the interfaces of each router using the topology shown above on Figure 2. For LoopBack (L0 and L1) interfaces use Class D addressing (255.255.255.255) and for all other interfaces use Class C (255.255.255.0). For the serial interfaces remember to set the clock rate to 56000 on the DCE side.

For example:
Router(config)# interface loopback0
Router(config-if)# ip address 192.1.1.1 255.255.255.255

STEP 2: A routing protocol must be configured to route the packets. Using OSPF enter the following commands on Router R1:

R1(config)# Router ospf 100
Network 192.168.11.1 0.0.0.0 area 0
Network 192.168.12.1 0.0.0.0 area 0
Your results should look similar to the window shown below:

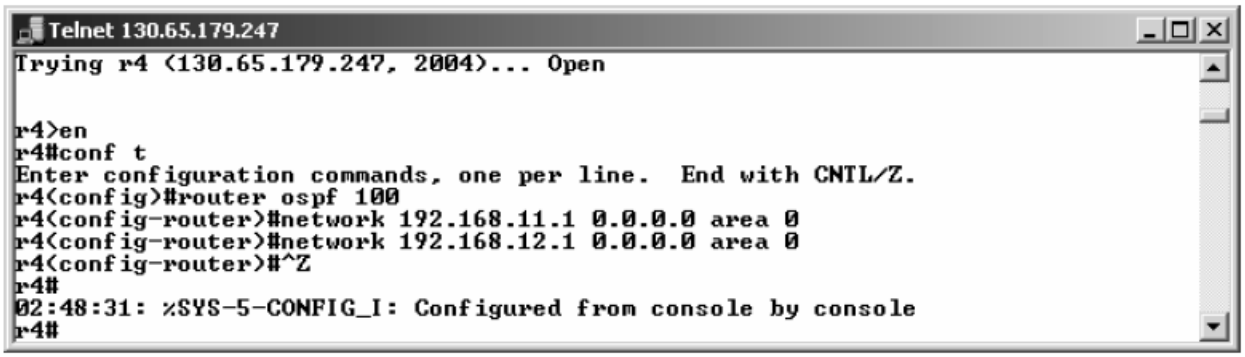


Figure 3 - R1 configuration of OSPF

STEP 3: To configure Router2 enter the following commands:
R2(config)# Router ospf *200*
Network 192.168.22.2 0.0.0.0 area 0
Network 192.168.12.2 0.0.0.0 area 0
Network 192.168.32.2 0.0.0.0 area 1
Network 192.168.23.1 0.0.0.0 area 1
Your results should look similar to the window shown below:

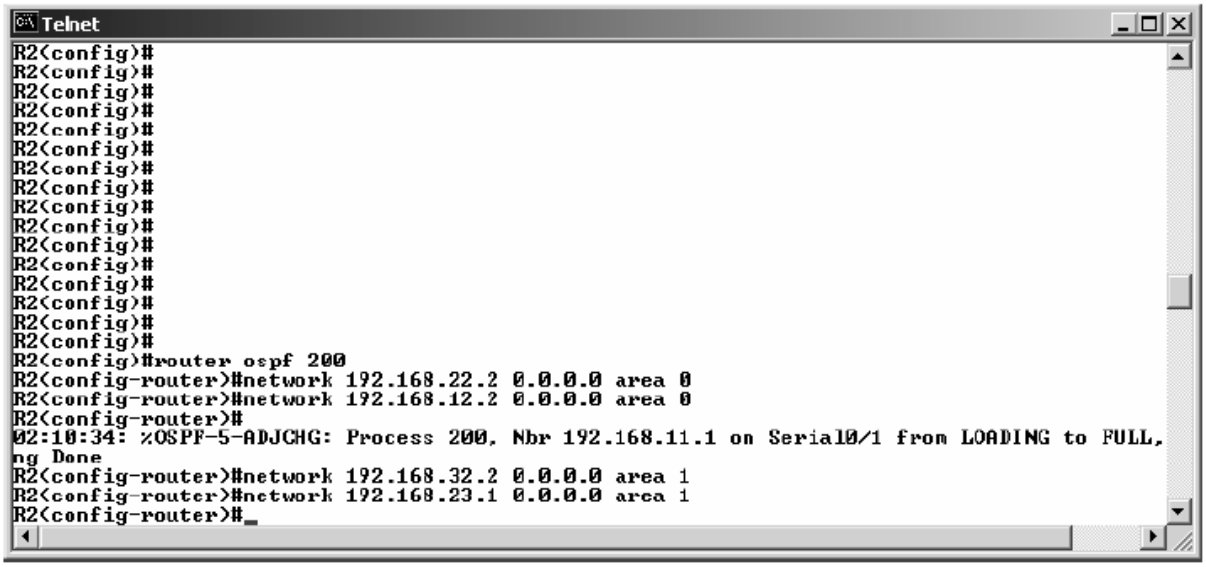


Figure 4 – R2 Configuration of OSPF

Important Points:
• Serial0/0 on Router1 and Serial0/1 on Router2 immediately go through the process of sending Hello Packets, forming a Neighbor Relationship. Once this relationship is formed, an *Adjacency* can be established, and the link state database information is exchanged.
• When the state of the Adjacency is *FULL* Router1's database is synchronized with its neighbor Router2. The different states of Adjacencies will be briefly discussed in the next section.

STEP 4: Configure OSPF on the interfaces of Router3.
Record the steps and commands that you entered. Make sure you put the correct interfaces into the correct areas based the given topology above.

Compare your results with the following window:

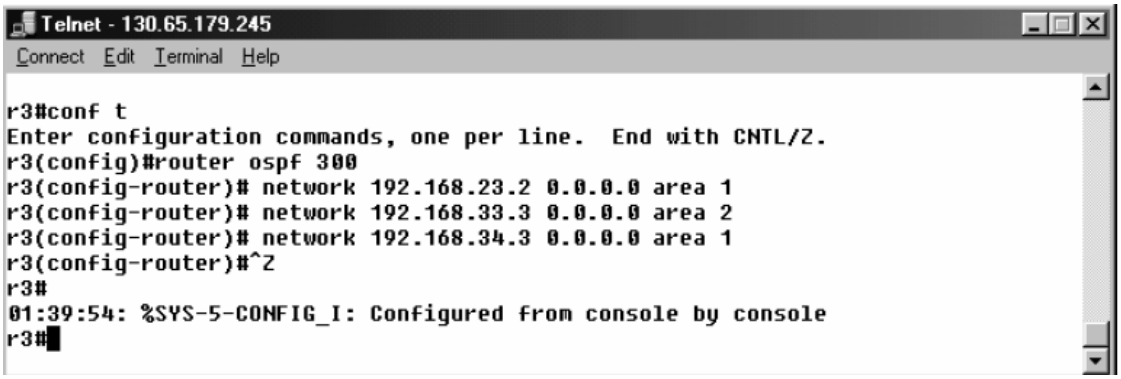


Figure 5 - R3 Configuration of OSPF

5. Verification of OSPF

To test the previous configuration perform the following steps.

EXERCISE 2: Verification of OSPF

STEP 1: To view the OSPF Neighbors create a session with R2 and use the command:
show ip ospf neighbor

Your results should be similar to the one in the window below:

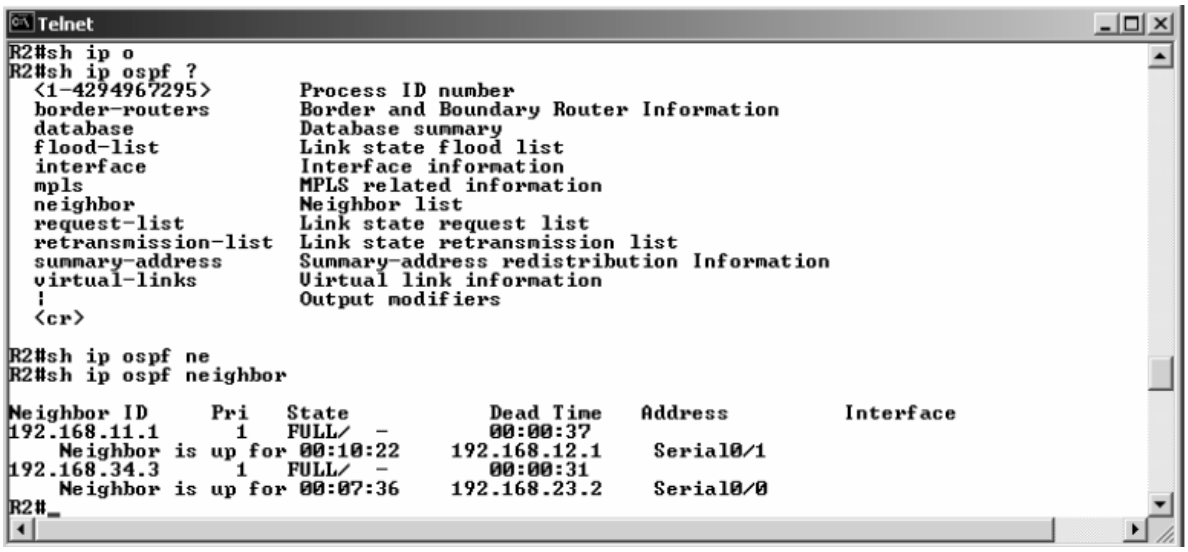


Figure 6 - show ip ospf neighbor on Router R2

What are the Router (neighbor) ID's of Router2's Neighbors?

What is the state of the adjacencies formed with the neighbors? (Note the "-" indicates no DR (Designated Router) or BDR (Backup Designated Router) for point-to-point interfaces.

The **Designated Router** becomes adjacent with all other routers on the network thereby reducing the number of adjacencies that need to be formed on that network. The **Backup Designated Router** backs up the DR should it fail.

What interfaces on R2 form the adjacencies to R1 and R3?

STEP 2: To check the OSPF interface on R2, enter:
show ip ospf interface

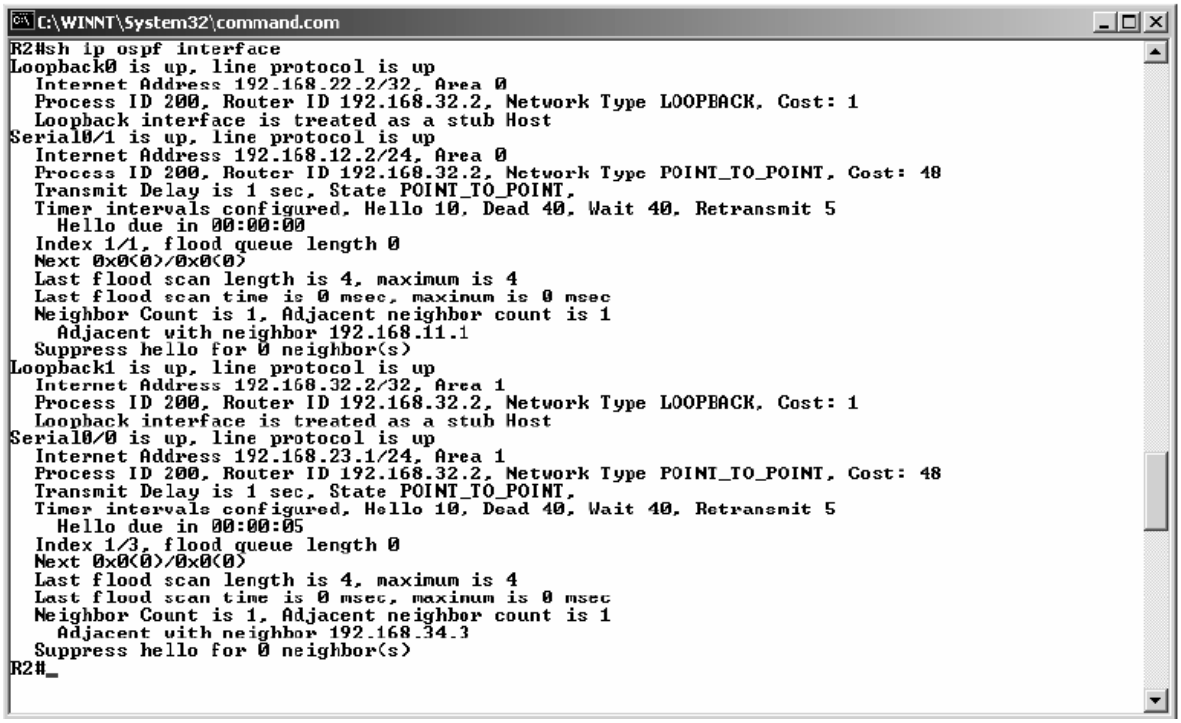


Figure 7 - show ip ospf interfaces on Router2

The output of this command shows the network type, timer intervals, and adjacent neighbors.

EXERCISE 3: Using Debug Commands to view Neighbors Set-Up

STEP 1: On R2 enable the debugging for OSPF events by entering the following command:
R2# debug ip OSPF events

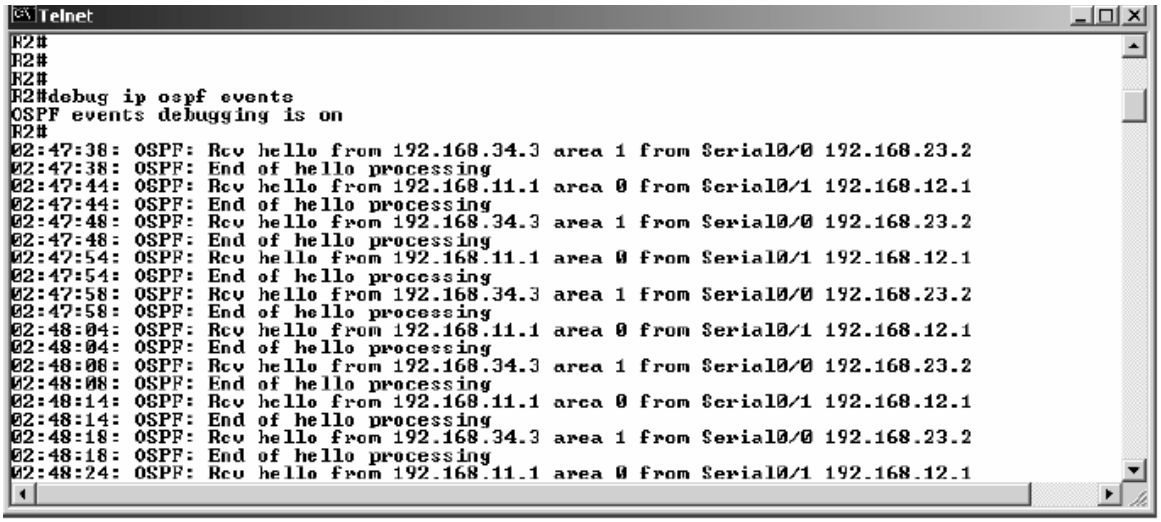


Figure 8 - debug ip ospf events on Router2

As shown, R2 is receiving Hello Messages from R1 and R3.

STEP 2: On R1 shut down Serial0/0. When you shut down this interface view the output displayed on *R2*. Observe how the change on the interface configuration of R2, as illustrated in the window below:

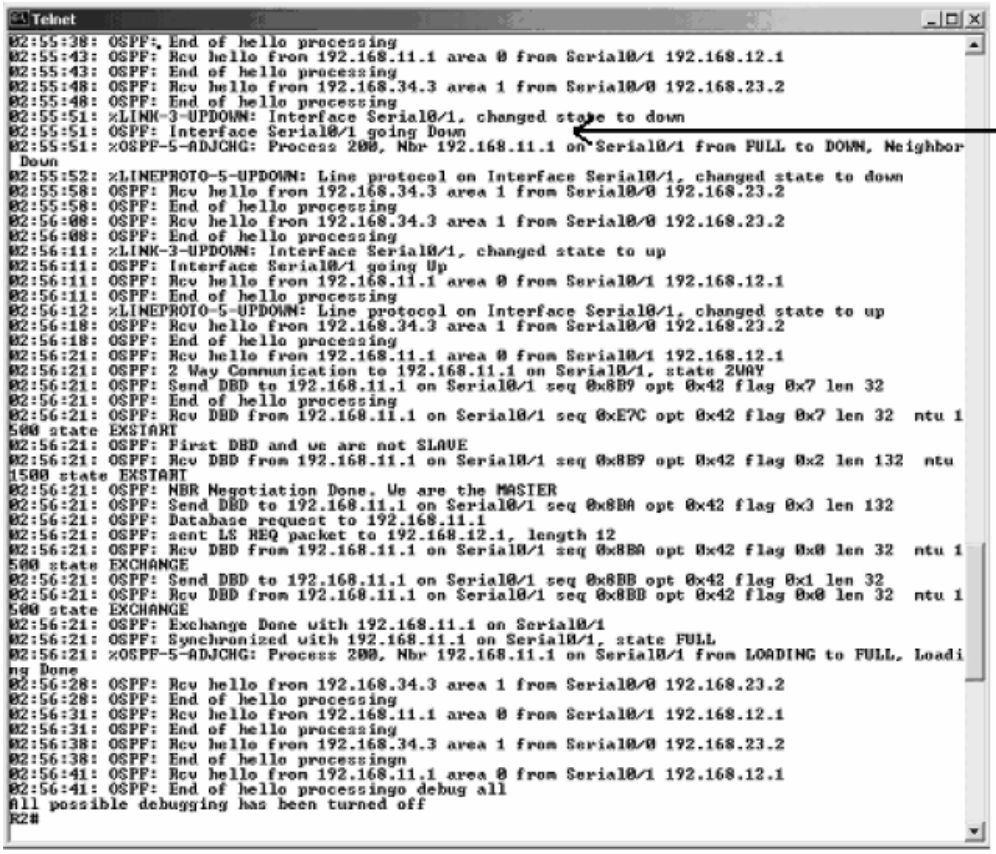


Figure 9 - Output Displayed on Router2 when Serial0/0 is shut down on Router1 and then brought back up with a no shut.

On Router2 you see that the adjacency with Router1 (ID 192.168.11.1) first changes to DOWN and the interface Serial0/1 on Router2 changes to DOWN as well.

STEP 3: Bring back up the interface Serial0/0 on Router1 with a no shutdown command. As seen in the above window the line protocol comes back UP. After the line comes up, the
two routers begin to establish a neighbor relationship once more. As you can see, a Master and Slave relationship is established where they agree on an initial sequence number. The sequence number is used to detect old or duplicate Link-State Advertisements (LSA). The stages are ExStart, Exchange, Loading, leading to the formation of a fully synchronized adjacency between Router1 and Router2.
• To turn off debugging on R2 enter **no debug ip ospf events**

EXERCISE 4: Viewing the routing tables and OSPF updates

STEP 1: To view the routing tables in each of the routers use the command:
show ip route
Your results should look similar to the window below:

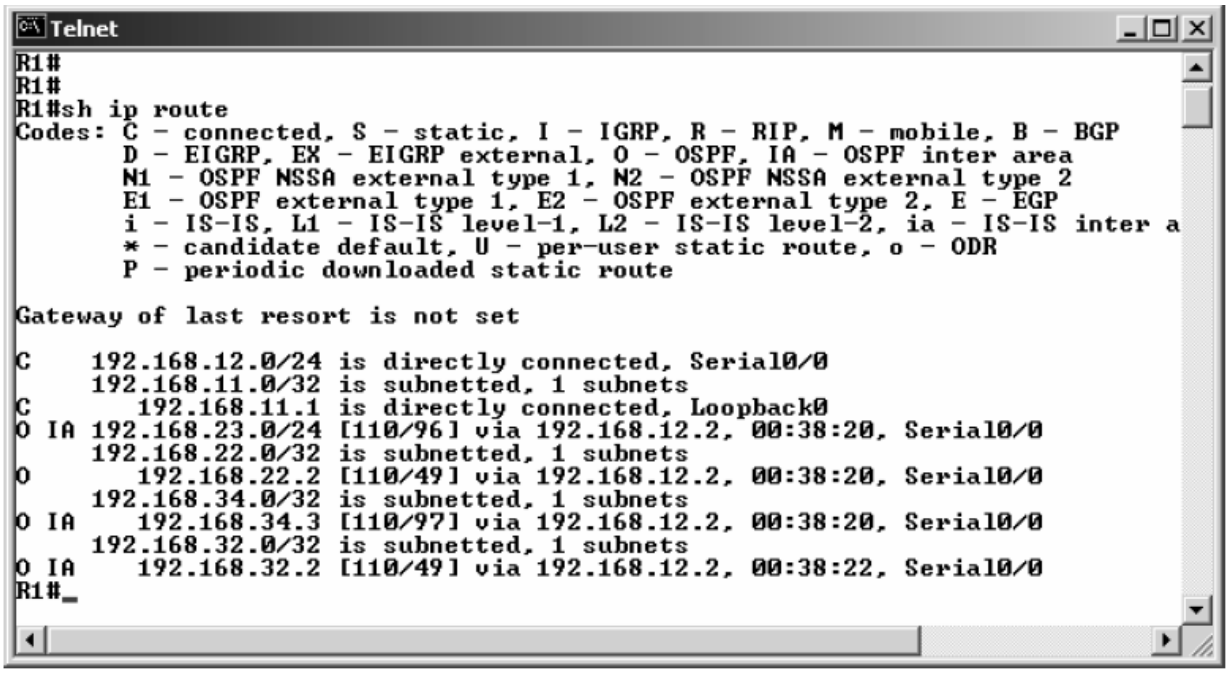


Figure 10 - Route Table of Router 1

Notice that some routes learned on Router1 are labeled with an O (OSPF derived protocol) and some with O IA (OSPF interarea route). What is the difference?

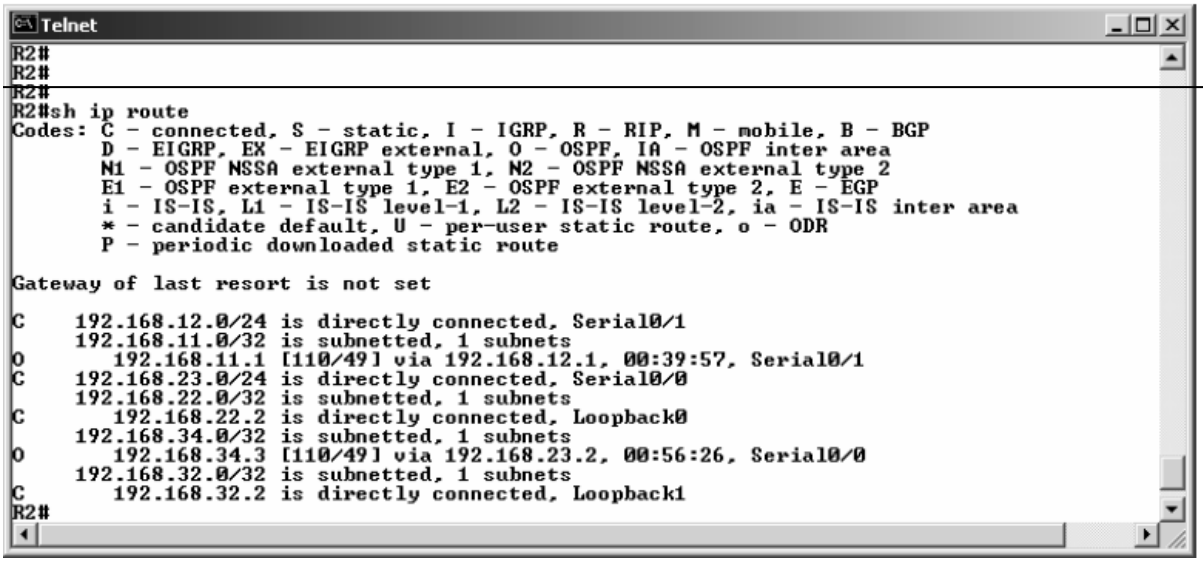


Figure 11 - Routing Table of Router 2

Looking at the routing table of R2 and the routing table from R1, is there a route not accounted for based on our topology?

Explain why the route is missing:

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a. Creating a Virtual Link

Since Area 2 is not directly connected to Area 0, the routers in Area 1 and Area 0 will not know about the loopback interface 192.168.33.3. This is because routers that are configured with OSPF send their LSA updates directly to Area 0. Area 0 then disseminates the updates to all directly connected areas.
For Area 0 to receive the LSA updates from Area 2 you need to create a Virtual Link connecting the two areas together. The transit Area will be Area 1, illustrated in the figure below:

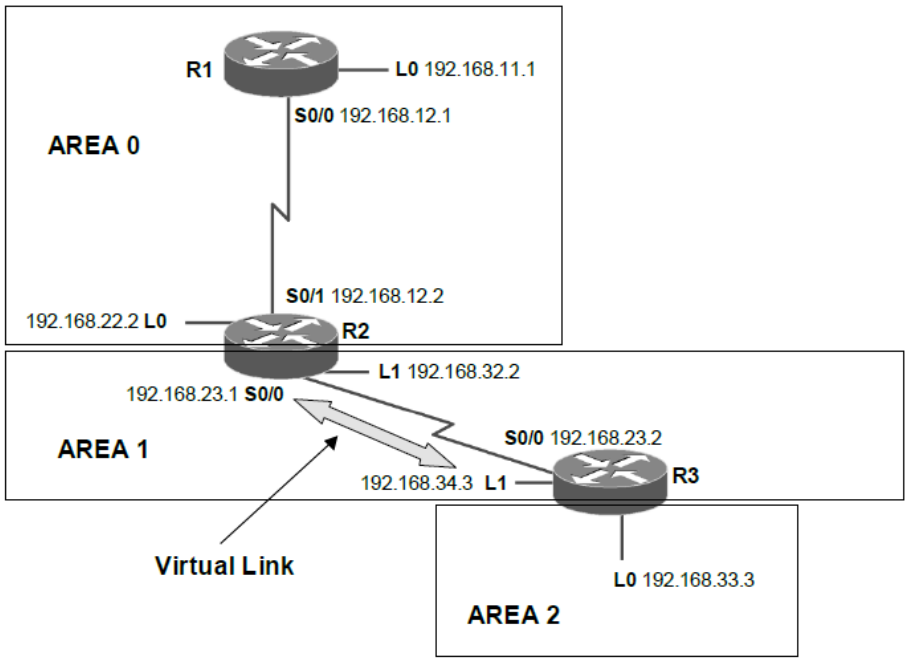


Figure 12 – Virtual Link with Area 1 as the Transit Area

To configure a Virtual Link you will be using the command:
area <*area id*> virtual link <*RID*>
Where "area id" is the transit area, and RID is the router ID. The router ID is the highest Loopback Address configured on the router. If you forgot the router ID of your neighbors, use the **show ip ospf neighbor** command.

EXERCISE 5: Configuring a Virtual Link

STEP 1: On Router 2 issue the following commands in the appropriate mode:
router ospf 200
area 1 virtual-link 192.168.34.3

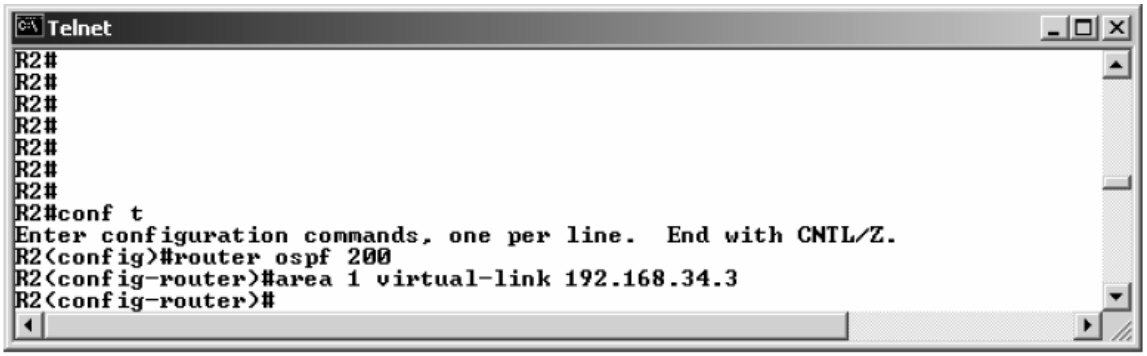


Figure 13 – Virtual Link Configured on Router2

STEP 2: On Router 3 configure the virtual link to Router 2.
Record your configurations below:

Check your configuration with the below graphic:

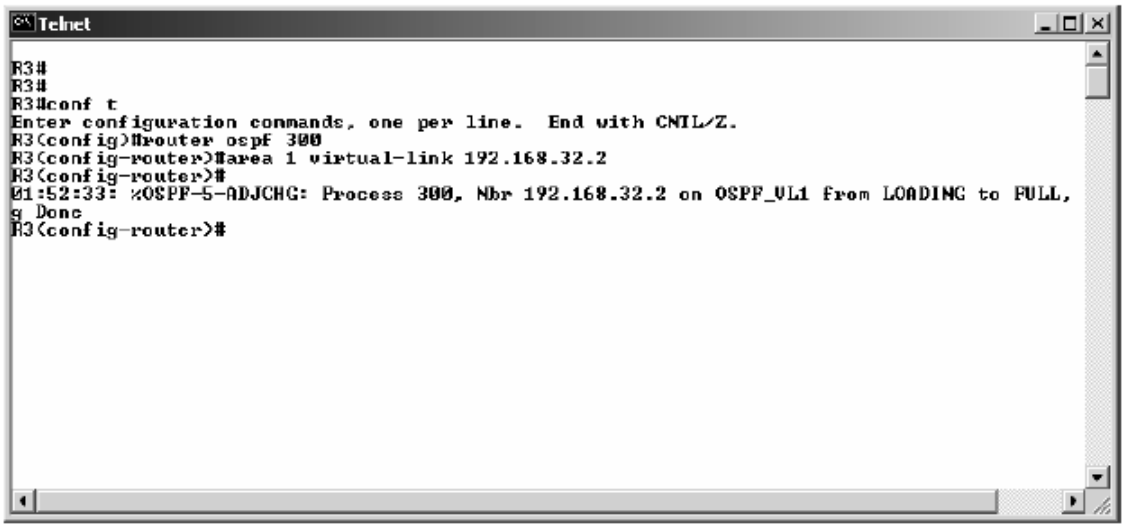


Figure 14 – Virtual Link Configured on Router3

STEP 3: Verify that the route to the loopback 192.168.33.3 on Router3 is learned by Router1 and Router2. View the routing table on Router1 and Router2 by typing **show ip route**.

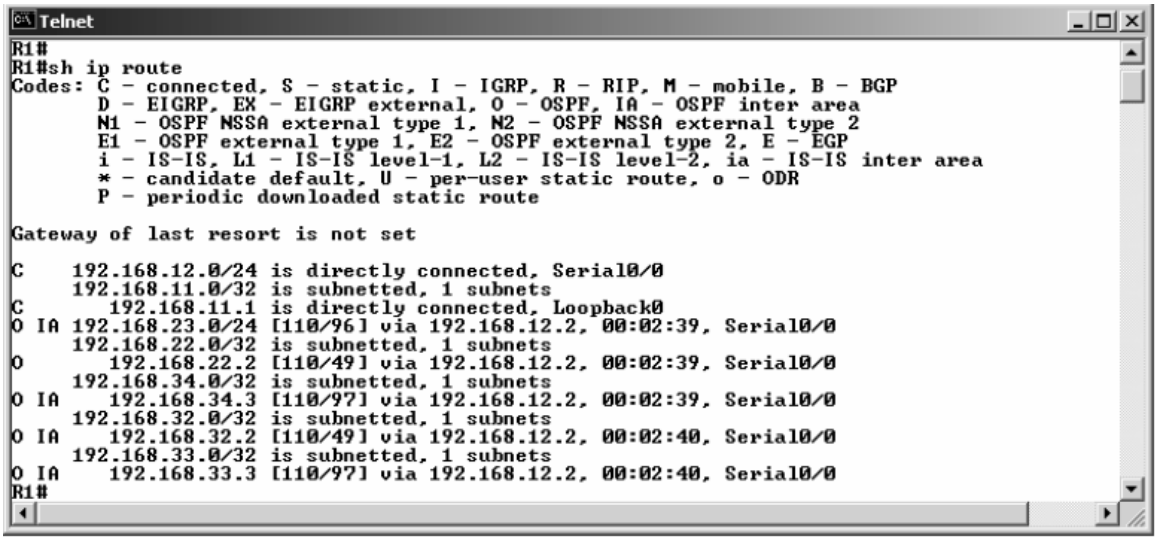


Figure 15 – Routing Table of Router1

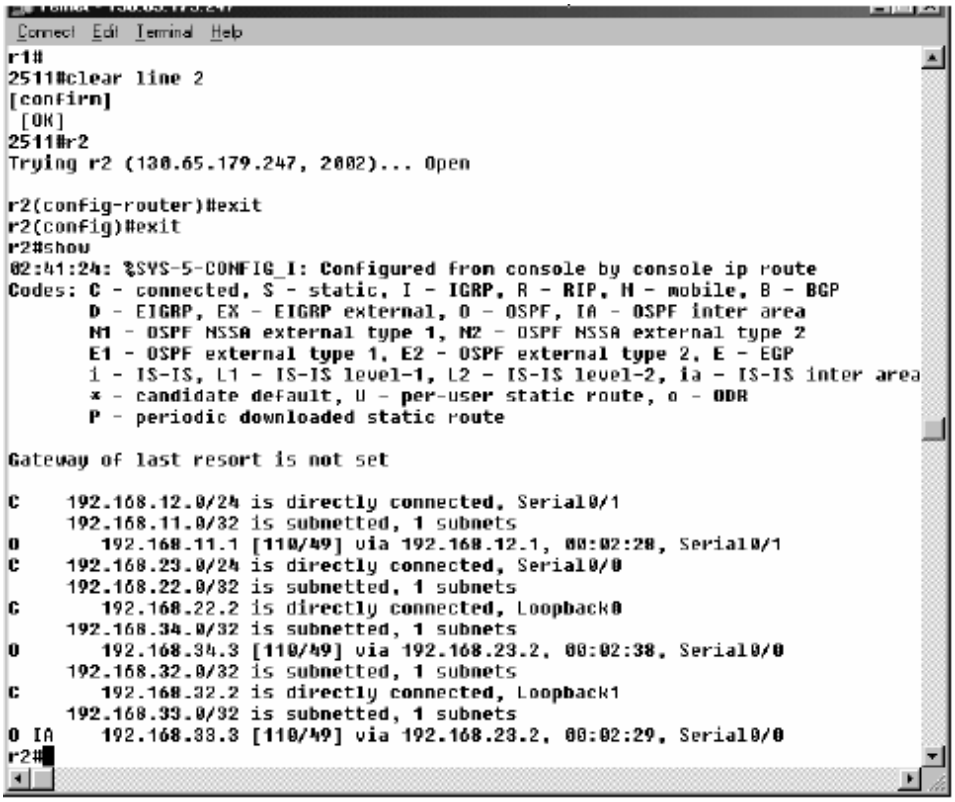


Figure 16 – Routing Table of Router2

STEP 4: Verify that the virtual-link is up using the following command:
show ip ospf virtual-links

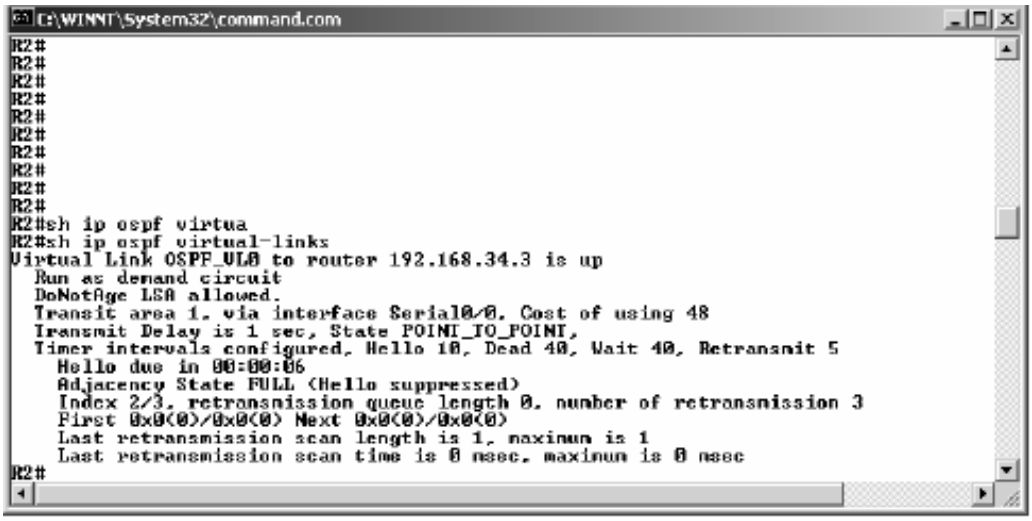


Figure 17 – show ip ospf virtual-link

EXERCISE 6: Configuring Network Address Translation

Background

NAT was originally intended to slow the depletion of available IP address space, by allowing a single public IP address to be used to represent many private IP addresses. NAT has since been used as a useful tool for network migrations and mergers, creating "virtual servers", and server load sharing. It is the reason we haven't run out of IPv4 addresses. NAT is typically configured on a border router to translate between local and global IP addresses.

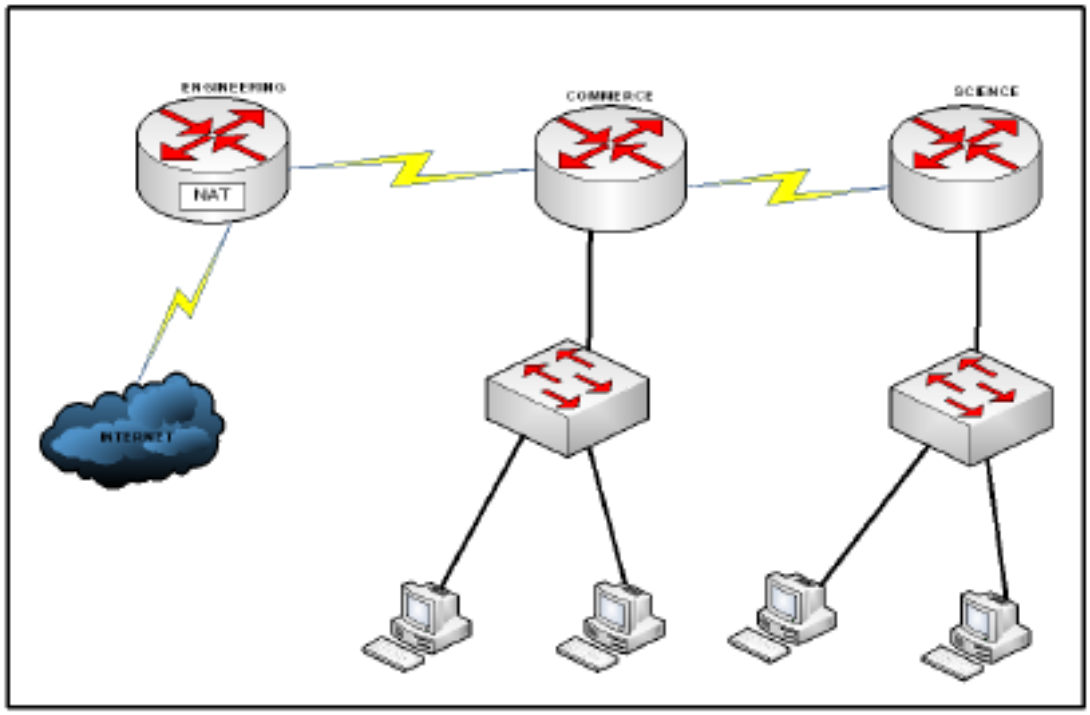


Figure 18: NAT configured at border router

Types of Network Address Translation

Static NAT: This type of NAT is designed to allow one to one mapping between local and global addresses. This requires you to have one real internet IP address for each host on your network.

Dynamic NAT: This gives you the ability to map an unregistered IP address to a registered IP address from out of a pool of registered IP addresses. You don't have to statically configure your router to map an inside to an outside address as you would with static NAT, but you do have to have enough real internet IP addresses for everyone who's going to be sending and receiving packets from the internet.

Overloading: This is the most popular type of NAT configuration. Understand that overloading really is a form of dynamic NAT that maps multiple unregistered IP addresses to a single registered IP address, many to one, by using different ports. It is also known as *Port Address Translation – PAT*. This allows you to have thousands of users connect to the internet using only one real global IP address. NAT overload is the reason we haven't run out of valid IP addresses on the internet.

Table 1 below summarises common NAT terminology.

Table 1: NAT Terminology

Names	**Meaning**
Inside local	Name of inside source address before translation
Outside local	Name of destination host before translation
Inside global	Name of inside host after translation
Outside global	Name of outside destination host after translation

Configuring Port Address Translation (PAT)

Port Address Translation (PAT) is the most commonly used form of NAT, also referred to as NAT overloading. It uses just one public IP address for all the users on a local network. Router 3 (R3) will act as the Internet Service Provider and Router 2 (R2) will therefore be the border device where the configuration of PAT will be done.

STEP 1: Create a NAT pool called **Access** on R2, with just one address. The one address in the pool, 192.168.23.50 will be "overloaded" by the private local addresses in the network. Enter the following command:
R2(config)#**ip nat pool Access 192.168.23.50 192.168.23.50 net 255.255.255.0**

STEP 2: Create access-list 1 on R2. This list permits traffic from the 192.168.12.0 network, and the other local IP addresses to be translated (loopback interfaces 192.168.11.1, 192.168.22.2, 192.168.32.2):

R2(config) # **access-list 1 permit 192.168.12.0 0.0.0.255**
R2(config)#**access-list 1 permit 192.168.11.1 0.0.0.0**
R2(config)#**access-list 1 permit 192.168.22.2 0.0.0.0**
R2(config)#**access-list 1 permit 192.168.32.2 0.0.0.0**

STEP 3: Map the access list to the pool that was created, allowing PAT to occur by using the **overload** command.
R2(config)#**ip nat inside source list 1 pool Access overload**

STEP 4: Configure serial 0/1 as an inside NAT interface
R2(config)#**interface serial 0/1**
R2(config-if)#**ip nat inside**

STEP 5: Configure serial 0/0 as an outside NAT interface
R2(config)#**interface serial 0/0**
R2(config-if)#**ip nat outside**

STEP 6: From R1, telnet to R3 which is the ISP router. Remember to add the **no login** command on R3 on the **vty 0 4** line, to allow you to telnet to R3.
R1#**telnet 192.168.23.2**

STEP 7: Execute the command **show users** from R3. (This shows who is accessing the VTY lines.)
R3#**show users**

a. What does it show as your source IP address? _____

b. What is your real source IP address? _____

The **show users** output should look something like this:

```
R3#show users
    Line       User       Host(s)              Idle       Location
*  0 con 0                 idle                 00:00:00
  67 vty 0                 idle                 00:00:29 192.168.23.50

  Interface    User               Mode         Idle     Peer Address
```

STEP 8: From R2, view your current translations by entering the **show ip nat translations** command.
R2#**show ip nat translations**
You should see something like this

```
R2#show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
tcp 192.168.23.50:1029 192.168.12.1:1029  192.168.23.2:23    192.168.23.2:23
```

a. What is the inside global address? _____

b. What is the outside global address? _____

STEP 9: On R2, turn on NAT debugging with the **debug ip nat** command.

R2#**debug ip nat**

STEP 10: From R1, ping R3. What are the translations shown for the source and destination addresses on R2?
R1#**ping 192.168.23.2**
_____
_____
_____
_____

Review

1.
To write a DOS command, which displays path taken by packets to reach www.cyberciti.biz,
you type: **tracert cyberciti.biz** in cmd

2.
Private networks can use IP addresses anywhere in the following ranges:

- 192.168.0.0 - 192.168.255.255 (65,536 IP addresses)
- 172.16.0.0 - 172.31.255.255 (1,048,576 IP addresses)
- 10.0.0.0 - 10.255.255.255 (16,777,216 IP addresses)

3.

To configure ip address of this type 192.168.35.2/28

11111111(255) 11111111(255) 11111111(255) 11110000(240)

You type: ip address 192.168.35.2 255.255.255.240

4.

To show only ip interfaces when troubleshooting you use the following command.

show ip interface brief